

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi r l$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

There are 54 questions in this paper.

The diagrams in this paper are not necessarily drawn to scale.

1. Evaluate $1.15 + 15$ correct to 3 significant figures.

A. 0076

B. 0077

C. 00766

D. 00767

E. 0076

2. $\frac{27^x}{3^y} =$

A. $\frac{9x}{y}$

B. $\frac{x}{9^y}$

C. 9^{x-y}

D. $\frac{3x}{3^y}$

E. 3^{3x-y}

3. Find the L.C.M. of $4x^2yz$ and $6xy^3$.

- A. $2xy$
- B. $12x^2y^3$
- C. $12x^2y^3z$
- D. $24x^2y^3z$
- E. $24x^3y^4z$

4. If $A = 2\pi r^2 + 2\pi rh$, then $h =$

- A. $A - r$
- B. $\frac{A}{r}$
- C. $\frac{A}{2\pi r} - r$
- D. $r - \frac{A}{2\pi r}$
- E. $\frac{A}{2\pi r} - 2\pi r^2$

5. Find the remainder when $x^3 - x^2 + 1$ is divided by $2x + 1$.

- A. -11
- B. $\frac{5}{8}$
- C. $\frac{7}{8}$
- D. $\frac{9}{8}$
- E. 5

6. Which of the following expressions has/have $b - c$ as a factor?

- I. $ab - ac$
- II. $a(b - c) - b + c$
- III. $a(b - c) - b - c$

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

7. Solve $1 < -3x + 4 < 10$.

- A. $-2 < x < 1$
- B. $-1 < x < 2$
- C. $x < -2$ or $x > 1$
- D. $x < -1$ or $x > 2$
- E. no solution

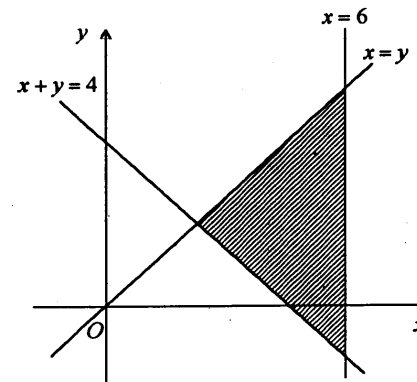
8. If $\frac{2}{x^2-1} \equiv \frac{a}{x+1} + \frac{b}{x-1}$, find a and b .

- A. $a = 2, b = 1$
- B. $a = 1, b = 2$
- C. $a = 1, b = 1$
- D. $a = 1, b = -1$
- E. $a = -1, b = 1$

9. In the figure, (x, y) is any point in the shaded region (including the boundary). Which of the following is/are true?

- I. $x \leq y$
- II. $x + y \leq 4$
- III. $x \leq 6$

- A. I only
- B. II only
- C. III only
- D. I and III only
- E. II and III only



10. Solve $\begin{cases} x^2 + y^2 = 13 \\ x + y = 1 \end{cases}$

- A. $\begin{cases} x = -2 \\ y = 3 \end{cases}$
- B. $\begin{cases} x = -6 \\ y = 7 \end{cases}$
- C. $\begin{cases} x = 2 \\ y = -1 \end{cases}$ or $\begin{cases} x = -3 \\ y = 4 \end{cases}$
- D. $\begin{cases} x = -2 \\ y = 3 \end{cases}$ or $\begin{cases} x = 3 \\ y = -2 \end{cases}$
- E. $\begin{cases} x = -6 \\ y = 7 \end{cases}$ or $\begin{cases} x = 7 \\ y = -6 \end{cases}$

11. If α and β are the roots of the equation $2x^2 + 4x - 3 = 0$, find $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

- A. $-\frac{22}{3}$
- B. $-\frac{16}{3}$
- C. $-\frac{14}{3}$
- D. $-\frac{8}{3}$
- E. $\frac{2}{3}$

12. Find the n -th term of the A.P. 4, 2, 0, -2,

- A. $2 + 2n$
- B. $4 - 2n$
- C. $4 + 2n$
- D. $6 - 2n$
- E. $(5 - n)n$

13. The sum to infinity of a G.P. is 2. If the first term is $\frac{3}{2}$, find the common ratio.

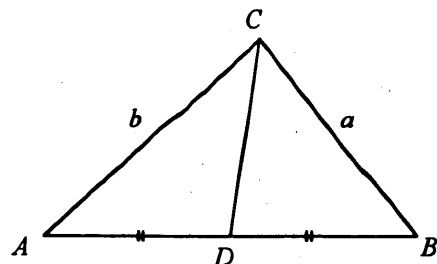
- A. $-\frac{1}{2}$
- B. $-\frac{1}{4}$
- C. $\frac{1}{4}$
- D. $\frac{1}{2}$
- E. $\frac{3}{2}$

14. Shop A offers a 10% discount on a book marked at $\$P$. Shop B offers a 15% discount on the same book marked at $\$Q$. If the selling price of the book is the same in both shops, express Q in terms of P .

- A. $Q = P + 5$
- B. $Q = \frac{17}{18}P$
- C. $Q = \frac{20}{21}P$
- D. $Q = \frac{21}{20}P$
- E. $Q = \frac{18}{17}P$

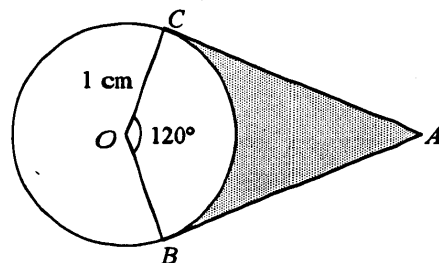
15. In the figure, area of $\triangle ACD$: area of $\triangle BCD$ =

- A. $1 : 1$.
 B. $a : b$.
 C. $b : a$.
 D. $a^2 : b^2$.
 E. $b^2 : a^2$.



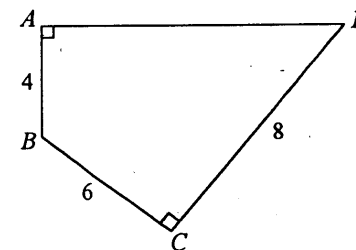
16. In the figure, O is the centre of the circle. AB and AC are tangents to the circle at B and C respectively. Area of the shaded region =

- A. $(2 - \frac{\pi}{6}) \text{ cm}^2$.
 B. $(2 - \frac{\pi}{3}) \text{ cm}^2$.
 C. $(\sqrt{3} - \frac{\pi}{6}) \text{ cm}^2$.
 D. $(\sqrt{3} - \frac{\pi}{3}) \text{ cm}^2$.
 E. $(\frac{\sqrt{3}}{2} - \frac{\pi}{6}) \text{ cm}^2$.



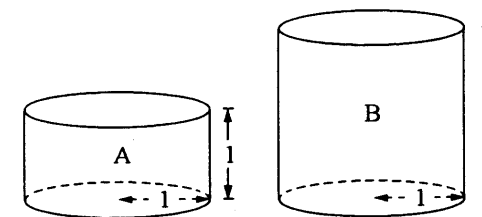
17. In the figure, the area of $ABCD$ is

- A. 36 .
 B. 40 .
 C. 44 .
 D. $4\sqrt{21} + 24$.
 E. $4\sqrt{29} + 24$.



18. In the figure, A and B are two right solid cylinders with the same base radius 1. If the heights of A and B are 1 and 2 respectively, find the total surface area of A
 the total surface area of B

- A. $\frac{1}{8}$
 B. $\frac{1}{4}$
 C. $\frac{1}{2}$
 D. $\frac{3}{5}$
 E. $\frac{2}{3}$



19. If $0^\circ \leq \theta \leq 360^\circ$, solve $2 \sin \theta = -\sqrt{3}$.

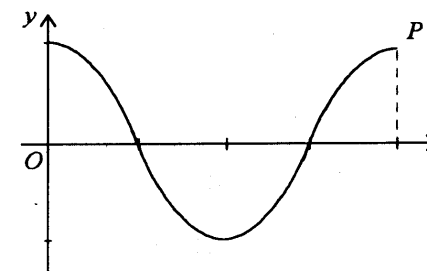
- A. 120° or 240°
- B. 120° or 300°
- C. 150° or 330°
- D. 210° or 330°
- E. 240° or 300°

20.
$$\frac{\frac{1}{\cos \theta} - \cos \theta}{\tan^2 \theta} =$$

- A. $\sin \theta$
- B. $\cos \theta$
- C. $\cos^2 \theta$
- D. $\frac{1}{\cos \theta}$
- E. $\frac{1}{\tan \theta}$

21. The figure shows the graph of $y = \frac{1}{2} \cos 2x$. The point P is

- A. $(\frac{\pi}{2}, 2)$
- B. $(\pi, \frac{1}{2})$
- C. $(\pi, 1)$
- D. $(2\pi, \frac{1}{2})$
- E. $(2\pi, 1)$



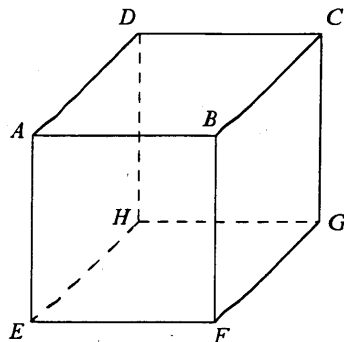
22. If $0 \leq x \leq \pi$, solve $2 \sin x + 3 \cos x = 0$ correct to 3 significant figures.

- A. 0.588
- B. 0.983
- C. 2.16
- D. 2.55
- E. no solution

23. The figure shows a cube. Which of the following is/are equal to $\angle AGE$?

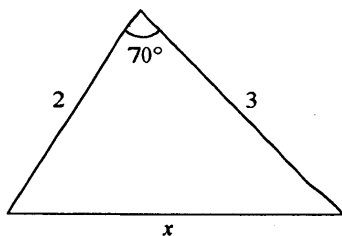
- I. $\angle AGF$
- II. $\angle BDF$
- III. $\angle DEG$

- A. I only.
- B. II only.
- C. III only.
- D. I and II only.
- E. II and III only.



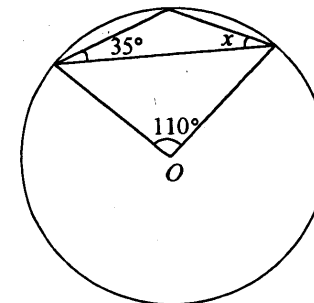
24. In the figure, find x correct to 3 significant figures.

- A. 2.71
- B. 2.98
- C. 3.31
- D. 3.88
- E. 4.14



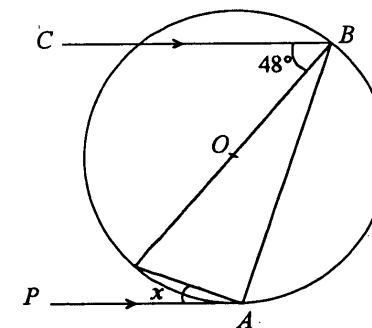
25. In the figure, O is the centre of the circle. Find x .

- A. 20°
- B. 27.5°
- C. 35°
- D. 37.5°
- E. 40°



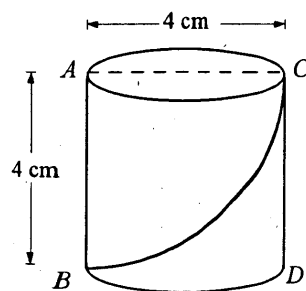
26. In the figure, O is the centre of the circle. PA is the tangent to the circle at A and $CB \parallel PA$. Find x .

- A. 21°
- B. 24°
- C. 42°
- D. 45°
- E. 48°



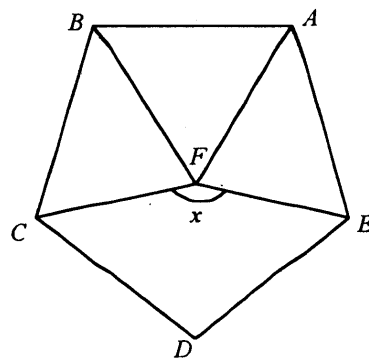
27. The figure shows a right circular cylinder with AC being a diameter of its upper face. AB and CD are two vertical lines on the curved surface. A curve is drawn on the surface of the cylinder from B to C . Find its shortest possible length.

- A. 2π cm
 B. $2\sqrt{\pi^2 + 4}$ cm
 C. $4\sqrt{2}$ cm
 D. $4\sqrt{\pi^2 + 1}$ cm
 E. $4\sqrt{\pi^2 + 4}$ cm



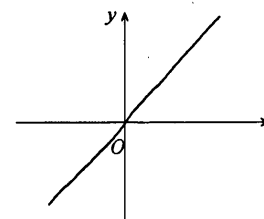
28. In the figure, $ABCDE$ is a regular pentagon and ABF is an equilateral triangle. Find x .

- A. 120°
 B. 126°
 C. 144°
 D. 156°
 E. 168°

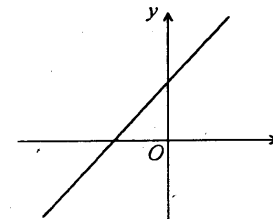


29. If a , b and c are all positive, which of the following may represent the graph of $ax + by + c = 0$?

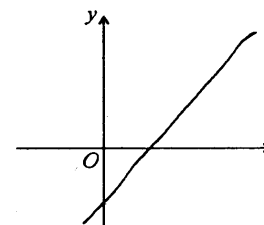
A.



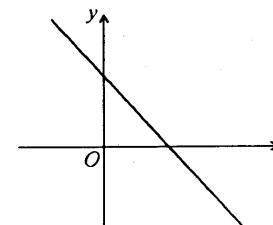
B.



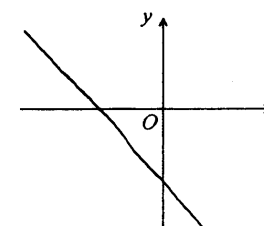
C.



D.



E.



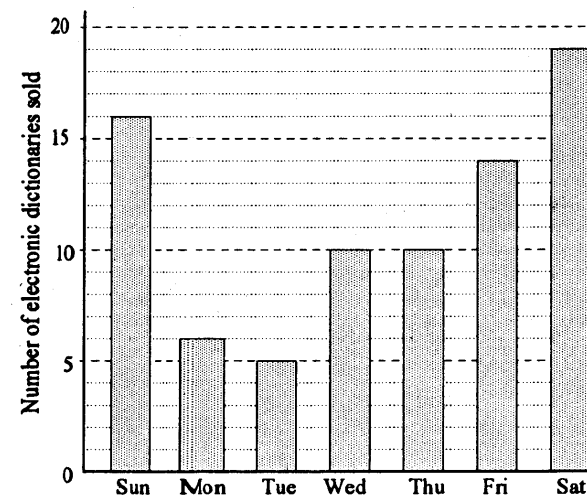
30. The equation of the circle centred at (a, b) and tangential to the x -axis is

- A. $x^2 + y^2 - 2ax - 2by + a^2 = 0$.
- B. $x^2 + y^2 - 2ax - 2by + b^2 = 0$.
- C. $x^2 + y^2 - 2ax - 2by + a^2 + b^2 = 0$.
- D. $x^2 + y^2 + 2ax + 2by + a^2 = 0$.
- E. $x^2 + y^2 + 2ax + 2by + b^2 = 0$.

31. Find the equation of the straight line which passes through $(3, -1)$ and is perpendicular to $2x - y + 1 = 0$.

- A. $x + 2y - 1 = 0$
- B. $x + 2y + 1 = 0$
- C. $x - 2y - 5 = 0$
- D. $2x + y - 5 = 0$
- E. $2x - y - 7 = 0$

32. The bar chart below shows the number of electronic dictionaries sold in a shop last week:



Of those electronic dictionaries sold last week, what percentage were sold on Sunday?

- A. 16%
- B. 18%
- C. 20%
- D. 22.5%
- E. 25%

33. Which of the following *cannot* be read directly from a cumulative frequency curve?

I. Mean
II. Median
III. Mode

- A. I only
B. II only
C. I and II only
D. I and III only
E. II and III only

34. There are 10 parcels. Two of them contain one pen each. If a man opens the parcels at random, what is the probability that he can find the two pens by opening two parcels only?

- A. $\frac{1}{25}$
B. $\frac{1}{45}$
C. $\frac{1}{50}$
D. $\frac{1}{90}$
E. $\frac{1}{100}$

35. In a certain game, the probability that John will win is 0.3. If he plays the game 3 times, find the probability that he will win at least once.

- A. 0.147
B. 0.441
C. 0.657
D. 0.9
E. 0.973

36. Simplify $\frac{1}{x-1} + \frac{1}{x+1} + \frac{3x-1}{1-x^2}$.

- A. $\frac{1}{1-x}$
B. $\frac{1}{1+x}$
C. $-\frac{1}{1+x}$
D. $\frac{3x+1}{1-x^2}$
E. $\frac{1-5x}{1-x^2}$

37. m and n are multiples of 3 and 4 respectively. Which of the following must be true?

I. mn is a multiple of 12.
II. The H.C.F. of m and n is even.
III. The L.C.M. of m and n is even.

- A. I only
B. I and II only
C. I and III only
D. II and III only
E. I, II and III

38. Let $x > y > 0$. If $\log(x+y) = a$ and $\log(x-y) = b$, then

$$\log \sqrt{x^2 - y^2} =$$

- A. $\frac{a+b}{2}$
B. $\frac{ab}{2}$
C. $\sqrt{a+b}$
D. $\sqrt{ab}$
E. $\sqrt{a} + \sqrt{b}$

39. If $\left(\frac{\sqrt{3}}{3} - \frac{1}{2}\right)x = 1$, then $x =$

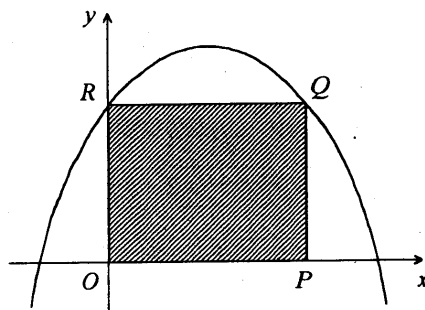
- A. $-\frac{\sqrt{3}}{3} + \frac{1}{2}$
B. $\frac{\sqrt{3}}{3} + \frac{1}{2}$
C. $-4\sqrt{3} - 6$
D. $4\sqrt{3} - 6$
E. $4\sqrt{3} + 6$

40. If 3 is a root of the equation $x^2 - x + c = 0$, solve $x^2 - x + c > 0$.

- A. $x < -2$ or $x > 3$
B. $x < 2$ or $x > 3$
C. $x > -6$
D. $-2 < x < 3$
E. $2 < x < 3$

41. The curve in the figure is the graph of $y = -x^2 + bx + c$. Find the area of the rectangle $OPQR$.

- A. bc
- B. b^2
- C. c^2
- D. $b^2 - 4c$
- E. $b^2 + 4c$



42. If the common difference of the A.P. $a_1, a_2, a_3, \dots$ is d , then the common difference of the A.P. $2a_1 + 3, 2a_2 + 3, 2a_3 + 3, \dots$ is

- A. 2.
- B. 3.
- C. d .
- D. $2d$.
- E. $2d + 3$.

43. The length of a rectangle is decreased by 20%. If the area remains unchanged, find the percentage increase of its width.

- A. $1\frac{1}{4}\%$
- B. $12\frac{1}{2}\%$
- C. $16\frac{2}{3}\%$
- D. 20%
- E. 25%

44. The following table shows the compositions of Tea A and Tea B which are mixtures of Chinese tea and Indian tea:

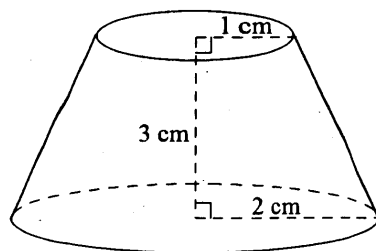
	Ratio of Chinese tea and Indian tea by weight
Tea A	3 : 1
Tea B	2 : 3

If 4 kg of tea A and 10 kg of tea B are mixed, find the ratio of Chinese tea and Indian tea in the mixture.

- A. 2 : 5
- B. 16 : 17
- C. 1 : 1
- D. 5 : 4
- E. 23 : 17

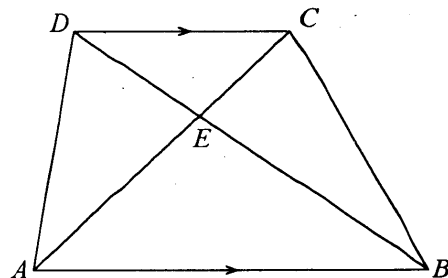
45. The figure shows a frustum of a right circular cone. The radii of the upper face and the base are 1 cm and 2 cm respectively. If the height is 3 cm, find the volume.

- A. $3\pi \text{ cm}^3$
 B. $\frac{9}{2}\pi \text{ cm}^3$
 C. $\frac{11}{2}\pi \text{ cm}^3$
 D. $7\pi \text{ cm}^3$
 E. $\frac{15}{2}\pi \text{ cm}^3$



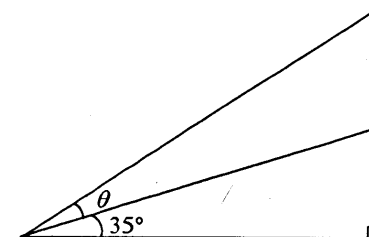
46. In the figure, if $\frac{\text{Area of triangle } CDE}{\text{Area of triangle } BCE} = \frac{1}{2}$,
 find $\frac{\text{Area of triangle } CDE}{\text{Area of trapezium } ABCD}$.

- A. $\frac{1}{10}$
 B. $\frac{1}{9}$
 C. $\frac{1}{8}$
 D. $\frac{1}{7}$
 E. $\frac{1}{6}$



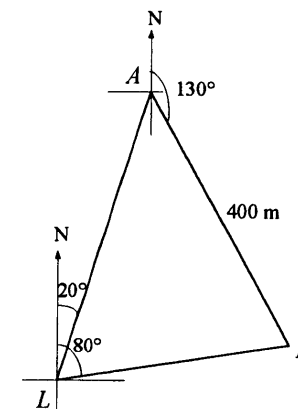
47. In the figure, find θ correct to the nearest degree.

- A. 16°
 B. 19°
 C. 26°
 D. 35°
 E. 36°



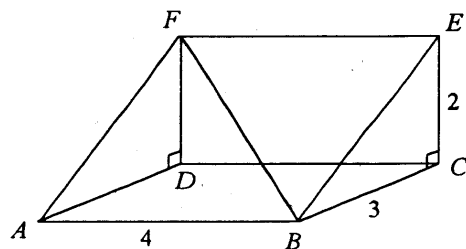
48. In the figure, the bearings of two ships A and B from a lighthouse L are 020° and 080° respectively. B is 400 m and at a bearing of 130° from A . Find the distance of B from L .

- A. 400 m
 B. $\frac{400}{\sin 60^\circ} \text{ m}$
 C. $\frac{400 \sin 50^\circ}{\sin 60^\circ} \text{ m}$
 D. $\frac{400 \sin 70^\circ}{\sin 60^\circ} \text{ m}$
 E. $\frac{400 \sin 70^\circ}{\sin 80^\circ} \text{ m}$



49. The figure shows a right prism with a right-angled triangle as the cross-section. Find the angle between the line BF and the plane $ABCD$ correct to the nearest degree.

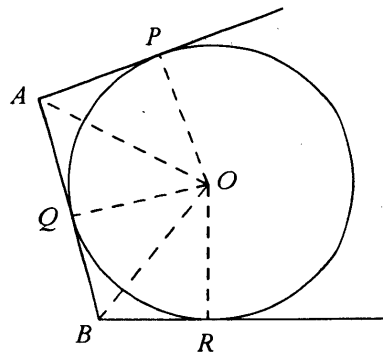
- A. 22°
 B. 34°
 C. 37°
 D. 42°
 E. 56°



50. In the figure, O is the centre of the circle. AP , AB and BR are tangents to the circle at P , Q and R respectively. Which of the following must be true?

- I. $AP + BR = AB$
 II. OQ bisects $\angle AOB$
 III. $\angle AOB = \frac{1}{2} \angle POR$

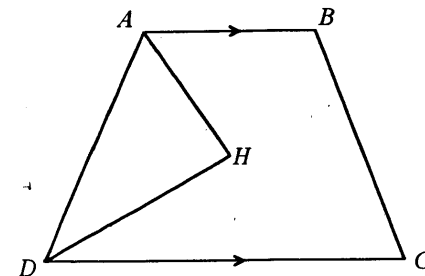
- A. I only
 B. II only
 C. I and II only
 D. I and III only
 E. I, II and III



51. In the figure, $ABCD$ is a trapezium with $AB \parallel DC$. AH bisects $\angle BAD$ and DH bisects $\angle ADC$. Which of the following must be true?

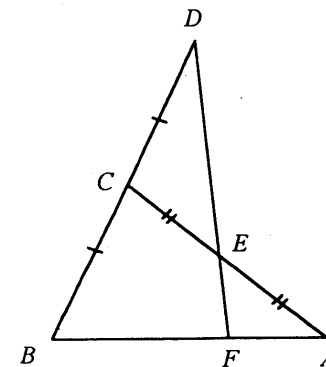
- I. $\angle AHD = 90^\circ$
 II. $\angle ADC = \angle BCD$
 III. $\angle BAD + \angle BCD = 180^\circ$

- A. I only
 B. II only
 C. III only
 D. I and III only
 E. II and III only



52. In the figure, $DE : EF =$

- A. $1 : 1$
 B. $2 : 1$
 C. $3 : 1$
 D. $3 : 2$
 E. $4 : 1$



53. $A(-3, 2)$ and $B(1, 3)$ are two points. C is a point on the AB produced such that $AB : BC = 1 : 2$. Find the coordinates of C .

A. $(-\frac{5}{3}, \frac{7}{3})$

B. $(-\frac{1}{3}, \frac{8}{3})$

C. $(3, \frac{7}{2})$

D. $(5, 4)$

E. $(9, 5)$

54. $C_1: x^2 + y^2 = 4$ and $C_2: x^2 + y^2 = 9$ are two circles. A chord AB of C_2 touches C_1 . Find the length of AB .

A. $\sqrt{5}$

B. $2\sqrt{5}$

C. $\sqrt{65}$

D. $2\sqrt{65}$

E. 10

END OF PAPER